

A PRE-REGISTERED PANEL OF ARTIFICIAL LISTENERS MEASURES PRONUNCIATION RECOVERABILITY

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ABSTRACT

Automatic pronunciation assessment depends on expert-scored speech. We treat heterogeneous speech recognizers as artificial listeners and measure how recoverable a known prompt is from a learner’s reading: each recognizer’s phoneme error against the prompt is a measurement, and the equal-weight mean over 28 recognizers admitted by an a-priori rule is the readout. Nothing is fitted to ratings. All design choices were fixed on the speechocean762 training split; five hypotheses were then pre-registered and tested once. The panel exceeds the best single recognizer on the held-out test split ($\Delta r = +0.038 [+0.021, +0.056]$), associates with segmental ratings far more than with rhythm or intonation on ERJ, replicates on EpaDB, correlates with human word recognition across 100 ALLSTAR talkers ($r = 0.576$), and tracks Mandarin tone ratings more than consonant ratings. High-error recognizers retain most of the signal, and no alternative aggregation rule improves on the mean. The measure is relative; absolute grades need labeled anchors.

Index Terms— pronunciation assessment, speech recognition, intelligibility, phoneme error rate, pre-registration

1. INTRODUCTION

Supervised assessors such as GOPT and HMamba learn multiple rating aspects from expert-scored speech [1, 2], and every new learner population, target language, or recording setting brings the same labeling cost [3]. We ask a measurement question instead of a prediction question: can the errors that a set of dissimilar recognizers make on a known prompt serve as a label-free measurement of how recoverable a learner’s pronunciation is, and what does that measurement correspond to in human judgment?

The idea rests on intelligibility research. Words are easier for some listeners to recover than for others [4], and intelligibility is a teaching target distinct from nativelikeness [5, 6]. Recognizers that differ in architecture, training data, and decoding are therefore probes with different failure modes. ASR-derived measures have long been associated with pronunciation ratings [7, 8], single-recognizer error features enter supervised assessors [9], and within-family Whisper

ensembles feed supervised intelligibility predictors [10, 11]. A recognizer that recovers accented speech almost perfectly, however, has little variation left with which to rank pronunciations [12]. Our instrument differs in three ways: the errors themselves are the measurement, the panel is defined by an inclusion rule rather than by performance, and nothing is fitted to ratings.

The contribution is a measurement instrument with a pre-registered validity study. Section 2 fixes the readout, the panel rule, seven hypotheses, and an unlock order for five corpora before any confirmatory data are read. Section 4 reports the development split, the five confirmatory tests, and secondary analyses of composition and aggregation. Code, inventories, and the pre-registration are public.¹

2. MEASUREMENT DESIGN

2.1. Readout

For utterance u with prompt t , listener k returns a hypothesis $h_k(u)$. Prompt and hypothesis pass through the same normalization (lowercasing, ASCII punctuation other than apostrophes to spaces, digits to spoken names) and the same `g2p_en` conversion to stress-stripped ARPAbet. The phoneme error rate $e_k(u) \in [0, 1]$ is the Levenshtein distance divided by the reference length, capped at one; an empty hypothesis scores one. Phone recognizers are compared directly with eSpeak reference phones, without inventory mapping. The readout is

$$p(u) = -\frac{1}{K} \sum_{k=1}^K e_k(u), \quad (1)$$

an equal-weight mean over the K admitted listeners; larger $p(u)$ means greater recoverability. The scale is relative, so evaluation uses correlation. For Mandarin, characters are simplified and pinyin conversion yields separate syllable-base and tone sequences, giving character, base, and tone error channels.

¹<https://github.com/mnaoizy/pal-pronunciation>

2.2. Panel rule

Admission is decided before any result is seen: openly downloadable weights; support for the target language or a universal phone inventory; one configuration per acoustic checkpoint, so decoder variants of an admitted checkpoint are excluded; no artificially degraded inputs; no configuration whose checkpoint identity is undocumented. Every configuration that passes is admitted, and none is removed for performance. Applied to the available English transcripts this admits 28 listeners in 10 families (Table 1); the Mandarin panel has 8.

Table 1. English panel by family with mean phoneme error on the speechocean762 training split. Moonshine returns empty output for 7.5% of those utterances, scored as full deletion.

GMM-HMM: sphinx-enus (0.65). **Kaldi hybrid:** vosk-small (0.33). **Convolutional CTC:** jasper (0.26), quartznet (0.26). **Self-supervised CTC, LibriSpeech:** wav2vec2-base (0.31), w2v2-large-lv60 (0.23), wav2vec2-robust (0.23), hubert-large (0.25), wavlm-large (0.30). **Self-supervised CTC, other domains:** w2v2-xlsr-cv (0.23), xlsr-1b-cv (0.25), w2v2-swbd (0.31), atc-w2v2 (0.35), atc-xlsr (0.56). **Whisper:** tiny (0.27), tiny.en (0.26), base.en (0.21), small.en (0.17), distil-small.en (0.18), distil-large-v3 (0.14), crisperwhisper (0.12). **Other encoder-decoders:** canary-1b (0.16), seamless-m4t-v2 (0.24), moonshine-base (0.27). **Transducer:** parakeet (0.15). **Speech LLM:** qwen2audio (0.13). **Phone recognizers:** espeak-phone (0.38), allosaurus-eng (0.70).

2.3. Hypotheses and protocol

Seven hypotheses were written before analysis. H1: the readout correlates with segmental pronunciation ratings. H2: the panel exceeds the best single listener, chosen on development data. H3: cross-family panels exceed single-family panels of equal size and matched strength. H4: the readout associates more with segmental than with prosodic ratings. H5: it correlates with human word recognition. H6: the tone channel tracks tone ratings more than consonant ratings. H7: high-error listeners alone retain signal comparable to the best single listener.

The speechocean762 training split (2,500 utterances, 125 speakers) was the only development data: readout, panel rule, statistics, and every comparator were fixed there. Confirmatory data were then unlocked once each in a fixed order, speechocean762 test, ERJ, EpaDB, ALLSTAR, OMPAL, each with one pre-specified primary comparison and a decision rule, after the pre-registration was committed to version control. Correlations are Pearson r with 2,000 speaker-clustered (talker-clustered for ALLSTAR) percentile bootstrap resamples, seed 20260913; paired differences use the same resamples. Secondary analyses are exploratory and uncorrected.

The comparators fixed on the training split are the single listener **hubert-large**, the 9 highest-error listeners (weak third), and 200 cross-family draws matched to the 7 Whisper listeners in mean error (± 0.03) and mean single-listener correlation (± 0.02).

3. DATA

speechocean762 [3] has 2,500 training and 2,500 test utterances from disjoint Mandarin-L1 speakers with five-rater consensus ratings of accuracy, completeness, fluency, prosody, and total. ERJ [13] has 3,800 sentences from 190 Japanese-L1 speakers, rated for segmental quality, rhythm, and intonation on disjoint utterances of the same speakers. EpaDB [14] has 3,160 utterances from 50 Argentinian-Spanish-L1 speakers with phrase scores (annotator 1). ALLSTAR [15, 16] supplies 1,500 clean HINT sentences from 100 talkers of 21 first languages; the criterion is each talker’s mean word-recognition accuracy by L1-English listeners across quiet and noise conditions. OMPAL [17] has 1,768 Mandarin utterances from 46 French-L1 learners with word-level tone and consonant ratings; its transcripts come from a re-decode that retained hypothesis text. All English transcripts were produced before this study; the transcript files, listener identities, row counts, and empty-output counts are catalogued in the released inventory.

4. RESULTS

4.1. Development split

On the training split the panel correlates with accuracy at $r = 0.735$ [0.678, 0.777], with total at 0.748, and with completeness at only 0.104. The best single listener, **hubert-large**, reaches 0.695. Random subsets of one, four, eight, and all 28 listeners average 0.604, 0.701, 0.721, and 0.735, so aggregation saturates within about ten members. Listener errors correlate 0.74 within families and 0.66 across families, and the first principal component carries 69% of their variance. The weak third alone reaches 0.714, above the strong third (0.689), and matched cross-family panels of seven exceed the seven Whisper listeners by +0.045 [+0.031, +0.060]. Figure 1 places every listener by error rate and rating correlation: recognition accuracy does not order the listeners as measurement probes, and the panel mean sits above all of them. These observations fixed the comparators; nothing else was changed.

4.2. Pre-registered confirmations

Table 2 lists the five primary comparisons. On the speechocean762 test split the panel reaches accuracy $r = 0.671$ [0.602, 0.725] against 0.633 for the fixed single listener, and 0.648, 0.663, 0.690, and 0.200 for fluency, prosody,

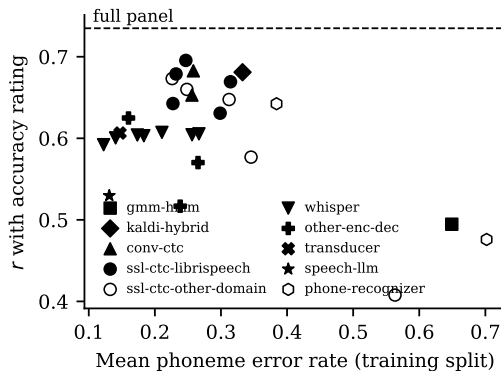


Fig. 1. Each admitted listener on the speechocean762 training split: mean phoneme error rate against correlation with the accuracy rating, by family. The dashed line is the equal-weight panel.

Table 2. Pre-registered primary comparisons, each run once on data unread during development (SO762 = speechocean762). Brackets: 2,000-resample clustered 95% intervals. All five decision rules (interval excludes zero) are met.

Data	Primary comparison	Result
SO762 test	H2: panel – best single, accuracy	+0.038 [+0.021, +0.056]
ERJ	H4: segmental – rhythm	+0.440 [+0.366, +0.516]
EpaDB	H1: r , phrase score	0.415 [0.301, 0.491]
ALLSSTAR	H5: talker r , word recovery	0.576 [0.404, 0.713]
OMPAL	H6: tone – consonant	+0.196 [+0.099, +0.293]

total, and completeness. For scale, the five raters agree pairwise at 0.618 on accuracy and each rater correlates with the mean of the other four at 0.732. Published supervised predictors report higher test accuracy on this benchmark (GOPT 0.720, HMamba 0.807 [1, 2]); they use its ratings for training, whereas the panel uses none. On ERJ the panel correlates with segmental quality at 0.597 [0.558, 0.632] but with rhythm at 0.157 and intonation at 0.073; the segmental-minus-intonation difference is +0.524 [+0.451, +0.597], and the speaker-level segmental correlation is 0.778. On EpaDB the panel exceeds hubert-large by +0.114 [+0.077, +0.152] and reaches 0.695 at the speaker level.

ALLSSTAR replaces expert ratings with the words that human listeners actually recovered (Fig. 2). The panel exceeds the fixed single listener (0.476) by +0.100 [+0.001, +0.200]; its margin over the seven Whisper listeners (0.548) is +0.028 [−0.001, +0.058]. On OMPAL the tone channel correlates with tone ratings at 0.389 [0.308, 0.459] and with consonant ratings at 0.192, whereas the syllable-base channel gives 0.257 and 0.270 and the character channel 0.370 and 0.245. Counting tone errors only on syllables whose base

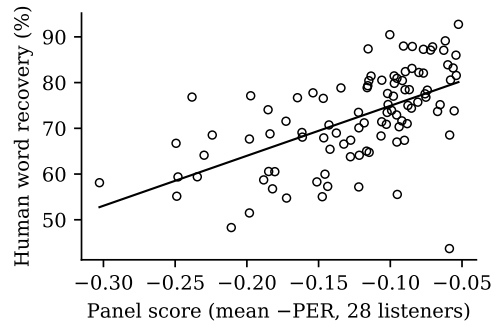


Fig. 2. Panel score and human word recognition for 100 ALLSSTAR talkers ($r = 0.576$ [0.404, 0.713]). Each point is a talker; the line is the least-squares fit.

pinyin matches the reference raises the tone correlation to 0.475 [0.402, 0.532] with consonant at 0.120, so the tone signal is not a by-product of base-syllable errors.

4.3. Composition and aggregation

Table 3. Secondary analyses on the speechocean762 test split (accuracy). Δ is paired against the full panel; the cross-family row is paired against the Whisper panel. Comparators were fixed on the training split.

Configuration	r	Δ [interval]
Full panel, equal-weight mean	0.671	—
Weak third (9 listeners)	0.649	+0.016* [−0.001, +0.034]
Whisper panel (7)	0.621	—
Matched cross-family, mean of 200 draws	0.649	+0.028 [+0.013, +0.042]
Median over listeners	0.652	−0.019 [−0.026, −0.012]
Minimum error per utterance	0.581	−0.089 [−0.154, −0.032]
Inverse training-error weights	0.666	−0.005 [−0.011, +0.001]

*paired against hubert-large.

Table 3 answers whether the instrument is more than an average of many recognizers. Every matched cross-family draw scores above the Whisper panel on the test split (100% of draws), but on ALLSSTAR the margin over Whisper includes zero, so family breadth is supported on the development benchmark and not established on the external criterion. The weak third alone reaches 0.649 [0.588, 0.700], within +0.016 of the best single listener, which is the pre-registered comparison for H7: weak recognizers carry signal because their errors fall on the segments that raters also penalize. Among aggregation rules over all 28 listeners, the median, the per-utterance minimum error, and inverse-training-error weights do not improve on the equal-weight mean. Random subsets of one, four, eight, and 28 listeners average 0.542, 0.637, 0.656, and 0.671, matching the development curve.

4.4. What the panel measures

Three properties follow from the confirmations. First, the readout is segmental: it tracks accuracy, segmental ERJ ratings, and word recovery, but rhythm and intonation only weakly, and its completeness correlation is low. Fluency and holistic ratings are reached only through the variance they share with accuracy. Second, the signal lives in errors that raters also penalize, not in recognition quality: the weakest third of the panel is as informative as the best single recognizer, while the most accurate recognizers, which recover accented speech almost regardless of pronunciation, contribute little variance. Third, aggregation matters more than curation: ten arbitrary listeners land within a few hundredths of the full panel, equal weights are not improved by medians or error-based weights, and the one composition effect that reached significance on the development split, family breadth, did not reach it on the external criterion. A practical recipe is therefore ten or more openly available recognizers of mixed origin, one configuration each, averaged without tuning; the measurement then transfers across learner L1s and to a human intelligibility criterion without any rating labels.

5. LIMITATIONS AND CONCLUSION

The readout is relative: mapping it to a grade requires a labeled calibration set. It has no time variable, so fluency and holistic ratings are reached only through the segmental correlation those ratings share with accuracy; a timing channel would be a separate instrument with its own validation. Phone recognizers are compared without inventory mapping, and the transcripts are retained outputs of unpinned model revisions, so bit-identical re-decoding is not claimed. Within these limits, an equal-weight panel of 28 recognizers admitted by rule measures prompt-conditioned recoverability without rating labels: it beats the best single recognizer on a held-out split, is segmental rather than prosodic, transfers to a new L1, tracks human word recovery, and extends to lexical tone, with each claim tested once under a committed protocol.

6. COMPLIANCE WITH ETHICAL STANDARDS

This study reanalyses existing speech corpora obtained through their providers' access procedures. No participants were recruited, and no new recordings or human ratings were collected. No separate ethical review was obtained for this secondary analysis.

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